

Lattice points inside non-lattice polygon

For lattice polygons there is Pick's formula to enumerate the lattice points inside the polygon. What about polygons with arbitrary vertices?

Let's process each of the polygon's edges individually, and after that we may sum up the amounts of lattice points under each edge considering its orientations to choose a sign (like in calculating the area of a polygon using trapezoids).

First of all we should note that if current edge has endpoints in $A=(x_1;y_1)$ and $B=(x_2;y_2)$ then it can be represented as a linear function:

$$y = y_1 + (y_2 - y_1) \cdot \frac{x - x_1}{x_2 - x_1} = \left(\frac{y_2 - y_1}{x_2 - x_1} \right) \cdot x + \left(\frac{y_1 x_2 - x_1 y_2}{x_2 - x_1} \right)$$

$$y = k \cdot x + b, \quad k = \frac{y_2 - y_1}{x_2 - x_1}, \quad b = \frac{y_1 x_2 - x_1 y_2}{x_2 - x_1}$$

Now we will perform a substitution $x = x' + \lceil x_1 \rceil$ so that $b' = b + k \cdot \lceil x_1 \rceil$. This allows us to work with $x_1' = 0$ and $x_2' = x_2 - \lceil x_1 \rceil$. Let's denote $n = \lfloor x_2' \rfloor$.

We will not sum up points at $x = n$ and on $y = 0$ for the integrity of the algorithm. They may be added manually afterwards. Thus we have to sum up $\sum_{x'=0}^{n-1} \lfloor k \cdot x' + b' \rfloor$. We also assume that $k' \geq 0$ and $b' \geq 0$. Otherwise one should substitute $x' = -t$ and add $\lceil b' \rceil$ to b' .

Let's discuss how we can evaluate a sum $\sum_{x=0}^{n-1} \lfloor k \cdot x + b \rfloor$. We have two cases:

- $k \geq 1$ or $b \geq 1$.

Then we should start with summing up points below $y = \lfloor k \rfloor \cdot x + \lfloor b \rfloor$. Their amount equals to

$$\sum_{x=0}^{n-1} \lfloor k \rfloor \cdot x + \lfloor b \rfloor = \frac{(\lfloor k \rfloor (n-1) + 2 \lfloor b \rfloor) n}{2}$$

Now we are interested only in points $(x;y)$ such that $\lfloor k \rfloor \cdot x + \lfloor b \rfloor < y \leq k \cdot x + b$. This amount is the same as the number of points such that $0 < y \leq (k - \lfloor k \rfloor) \cdot x + (b - \lfloor b \rfloor)$. So we reduced our problem to $k' = k - \lfloor k \rfloor$, $b' = b - \lfloor b \rfloor$ and both k' and b' less than 1